Paired tracking settings: quantitative comparison

2024 GroupTrial | six beetles | five source TOMLs | background subtraction ON in both arms

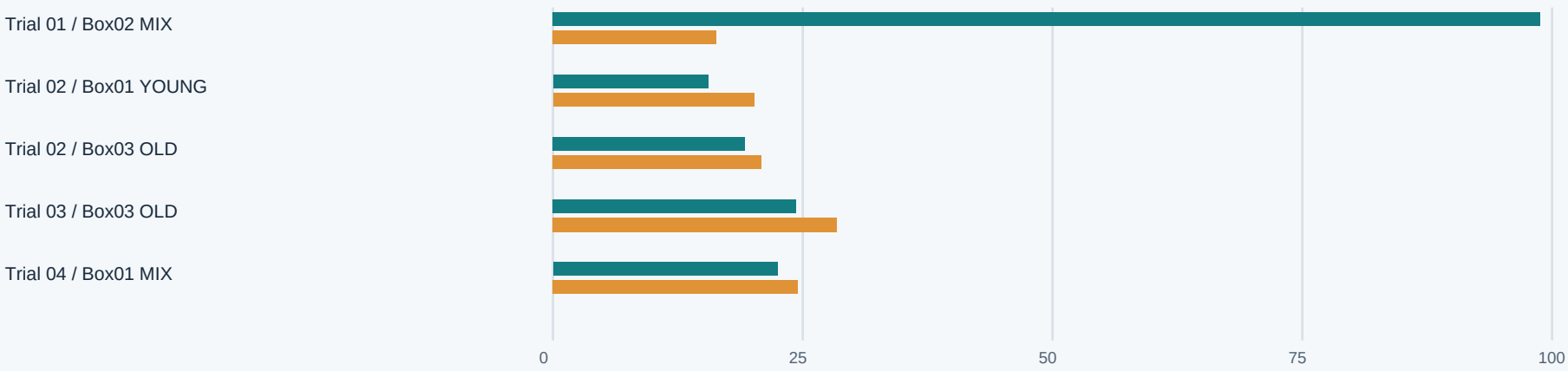
Human settings retain each source's original intensity and blob area; shared settings use 50-255 and 400 px2.
Original start/end frames and all four ROIs are identical within each pair. End frame is exclusive.

5 / 5 pairs measured

Source / original settings	Missing: human	Missing: shared	Change (pp)	Est. ID H / C	>100px H / C
Trial 01 / Box02 MIX Start 422 | 0-143 / 329 px2	98.8%	16.4%	-82.4	96.7 / 95.3	0 / 1
Trial 02 / Box01 YOUNG Start 242 | 43-255 / 228 px2	15.5%	20.1%	+4.6	94.2 / 94.1	5 / 8
Trial 02 / Box03 OLD Start 484 | 31-255 / 286 px2	19.2%	20.9%	+1.7	99.7 / 93.7	0 / 0
Trial 03 / Box03 OLD Start 420 | 32-255 / 171 px2	24.3%	28.4%	+4.1	99.1 / 99.8	1 / 0
Trial 04 / Box01 MIX Start 175 | 55-255 / 56 px2	22.5%	24.5%	+2.1	90.1 / 95.9	15 / 4

Missing-coordinate rate by source

Bars use the same 0-100% scale. Lower is better for missingness; an identity estimate is not visual proof.



IDtracker.ai output measures; identity continuity and presumed social contact require visual review.

Which beetles changed?

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Each cell is shared minus human missing-coordinate percentage points. Green means fewer missing cells; orange means more.
Beetle IDs below are zero-based IDs from the raw trajectory array (0-5).

	ID 0	ID 1	ID 2	ID 3	ID 4	ID 5
Trial 01 / Box02 MIX Human 98.8% | Shared 16.4%	-89.8	-59.5	-92.7	-66.8	-96.7	-89.0
Trial 02 / Box01 YOUNG Human 15.5% | Shared 20.1%	+9.3	-3.6	+2.0	+17.5	+2.0	+0.4
Trial 02 / Box03 OLD Human 19.2% | Shared 20.9%	+12.0	+2.3	+0.0	-5.5	+0.0	+1.2
Trial 03 / Box03 OLD Human 24.3% | Shared 28.4%	+45.8	-27.9	+0.0	+9.5	-42.9	+40.0
Trial 04 / Box01 MIX Human 22.5% | Shared 24.5%	+3.1	+17.4	+0.9	+0.4	-1.6	-8.0

Interpretation

Compare within each source only. Very sparse outputs can still have a high IDtracker estimated accuracy.
Jumps count adjacent raw assigned coordinates >100 px, not movements reconstructed by the pile model.

IDtracker.ai output measures; identity continuity and presumed social contact require visual review.